

Model Risk Lab

What a Wrong Pricing Model Costs an Options Desk — in Dollars

A complete, from-scratch account: every model, every estimator, every test, every observation.
Calibrated to real SPY and NIFTY 50 market data.

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Abstract

A market maker earns the bid–ask spread by quoting around *fair value* — but fair value comes from a model, and the model can be wrong. This document is a self-contained account of a project that measures the cost of that error in dollars and then detects it statistically. It is written for a reader with bachelor’s-level mathematics and statistics and assumes nothing beyond that: Brownian motion, the Poisson process, Itô’s lemma, option pricing, maximum-likelihood theory, the likelihood-ratio/AIC/BIC machinery, Value-at-Risk and Expected Shortfall, the Kupiec and Christoffersen backtests, the bootstrap and the CUSUM monitor are each built up from first principles before they are used. Headline findings: on 17+ years of SPY and NIFTY 50 daily returns the Gaussian (Black–Scholes) model is rejected against the Merton jump-diffusion with likelihood-ratio statistics of ≈ 1627 and ≈ 1313 ; on 16 March 2020 the SPY options market priced 3.7 crashes per year of average size -31% ; a simulated desk that prices with the matched model realises $+0.153$ per contract against a promised 0.15 (bootstrap 95% CI $[0.146, 0.159]$), while the same desk pricing Black–Scholes in the jump world *believes* $+0.15$ and realises -10.20 ; no half-spread below \$15 makes the wrong-model desk 95%-safe, versus 8.8 cents for the right one; and a validation suite catches the broken model at trade 83 of 1,467 with zero false alarms on the healthy desk.

Contents

I	Foundations	4
1	Random variables, moments, and why tails matter	4
1.1	Expectation, variance, and higher moments	4
1.2	The normal distribution and the size of its tails	4
2	Brownian motion	4
2.1	From coin flips to a continuous process	4
2.2	The one strange rule: $(dW)^2 = dt$	5
2.3	Itô’s lemma	5
3	The Poisson process	5
II	Models of Price Motion and Option Pricing	5
4	Geometric Brownian motion: the Black–Scholes world	6
4.1	The model	6
4.2	Solving it, and the origin of $-\frac{1}{2}\sigma^2$	6
4.3	What GBM asserts about the world	6

5	The Merton jump-diffusion: a world that can gap	6
5.1	Construction	6
5.2	The distribution of a daily return: a Poisson mixture of Gaussians	7
6	Options and arbitrage-free pricing	7
6.1	Contracts and payoffs	7
6.2	No-arbitrage and put–call parity	7
6.3	Delta hedging and the Black–Scholes argument	7
6.4	Risk-neutral pricing in one paragraph	8
7	Pricing under Merton: a series of Black–Scholes prices	8
7.1	Implied volatility and the smile	8
III	Estimation: Fitting the Models to Real Data	9
8	Statistical machinery from scratch	9
8.1	Likelihood and maximum likelihood	9
8.2	The delta method	9
8.3	Hypothesis tests, p-values, and the likelihood-ratio test	9
8.4	AIC and BIC	10
8.5	QQ plots	10
9	Calibration I: the physical measure (what returns actually do)	10
9.1	GBM: the MLE in closed form	10
9.2	Merton: numerical MLE on the mixture density	10
9.3	Results and observations	11
10	Calibration II: the risk-neutral measure (what options cost)	11
10.1	Why fit option prices separately	11
10.2	The fitting problem, in detail	12
10.3	Results and observations	12
IV	The Desk: Market-Making P/L Under Model Risk	13
11	The simulation, piece by piece	14
11.1	The quoting cycle	14
11.2	Why a hedged option earns “premium minus fair value”	15
11.3	Client flow: elastic, informed, put-heavy	15
11.4	The four scenarios	16
12	Results I: what happened, and why	16
V	Risk Measurement: Pricing the Damage	17
13	VaR and Expected Shortfall, from scratch	17
14	The model-risk premium: a break-even spread search	18
VI	Validation: Catching the Wrong Model from P/L Alone	18

15 VaR backtesting	19
15.1 The desk forecasts its own risk	19
15.2 Kupiec's proportion-of-failures test, derived	19
15.3 Christoffersen's independence and conditional-coverage tests	20
16 The bootstrap: honest error bars on realised edge	21
17 CUSUM: the early-warning monitor	21
17.1 The problem: detect a change, fast, without crying wolf	21
17.2 The CUSUM statistic	21
VII Closing Material	22
18 Limitations, stated plainly	22
19 Reproducibility and repository map	23

Part I

Foundations

The whole project rests on a small set of probabilistic and statistical ideas. This part builds them from scratch so that nothing later has to be taken on faith.

1 Random variables, moments, and why tails matter

1.1 Expectation, variance, and higher moments

A random variable X has expectation (mean) $\mathbb{E}[X]$, the probability-weighted average of its outcomes, and variance

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2], \quad \text{sd}(X) = \sigma_X = \sqrt{\text{Var}(X)}. \quad (1)$$

Mean and variance describe location and spread, but two distributions can share both and still be utterly different *in the tails* — and finance lives and dies in the tails. Two standardised moments capture the difference. Writing $Z = (X - \mathbb{E}X)/\sigma_X$:

$$\text{skewness} = \mathbb{E}[Z^3], \quad \text{excess kurtosis} = \mathbb{E}[Z^4] - 3. \quad (2)$$

Skewness measures asymmetry (negative skew: the long tail is on the loss side). Excess kurtosis measures tail weight relative to the normal distribution, whose $\mathbb{E}[Z^4]$ is exactly 3 — hence the “−3”. Daily equity returns typically show negative skew and large positive excess kurtosis; a Gaussian has zero of both.

1.2 The normal distribution and the size of its tails

The normal (Gaussian) law $\mathcal{N}(\mu, \sigma^2)$ has density

$$\varphi(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]. \quad (3)$$

It appears everywhere because of the **central limit theorem**: sums of many small independent shocks are approximately Gaussian. That is exactly the logic behind Gaussian models of returns — a day’s return is the sum of thousands of small trades. The logic fails when *one* shock is occasionally enormous (a crash): then the sum is dominated by a single term and the CLT approximation breaks in the tails. How thin are Gaussian tails? For $Z \sim \mathcal{N}(0, 1)$:

k	2	3	5	7	9
$\mathbb{P}(Z < -k)$	2.3×10^{-2}	1.3×10^{-3}	2.9×10^{-7}	1.3×10^{-12}	1.1×10^{-19}
“once every”	2 months	3 years	13,000 yrs	3×10^9 yrs	3×10^{16} yrs

(“once every” converts the probability to trading days.) Keep this table in mind: Section 9.3 shows SPY’s worst in-sample day is a $\approx 9\sigma$ event under the fitted Gaussian — an event that model says should not happen in the age of the universe, yet it is right there in the data.

2 Brownian motion

2.1 From coin flips to a continuous process

Take a symmetric random walk: every time step Δt , move up or down by $\sqrt{\Delta t}$ with probability $\frac{1}{2}$ each. Over time t there are $t/\Delta t$ steps, so the position has mean 0 and variance $(t/\Delta t) \cdot \Delta t = t$. Letting $\Delta t \rightarrow 0$, the CLT turns the position at time t into a Gaussian $\mathcal{N}(0, t)$, and the limiting object is **Brownian motion** W_t , characterised by:

1. $W_0 = 0$;
2. independent increments: $W_t - W_s$ is independent of everything up to s ;
3. Gaussian increments: $W_t - W_s \sim \mathcal{N}(0, t - s)$;
4. continuous paths.

2.2 The one strange rule: $(dW)^2 = dt$

Over a short interval dt the increment dW has standard deviation $\sqrt{dt}$, which is *much larger* than dt itself when dt is small. Consequently the squared increment $(dW)^2$ has mean dt and negligible variance around it: to first order in dt ,

$$(dW)^2 = dt \quad (\text{the quadratic-variation rule}). \quad (4)$$

This is the single fact that makes stochastic calculus different from ordinary calculus, and it is the origin of every “ $\frac{1}{2}\sigma^2$ ” term below.

2.3 Itô’s lemma

Let X_t follow $dX = a dt + b dW$ and let $f(x)$ be smooth. A Taylor expansion of $f(X_{t+dt})$ keeps terms up to second order, because by (4) the second-order term $(dX)^2 = b^2 dt$ is *first order in time*:

$$df = f'(X) dX + \frac{1}{2}f''(X) (dX)^2 = \left(af'(X) + \frac{1}{2}b^2f''(X)\right) dt + bf'(X) dW. \quad (5)$$

This is **Itô’s lemma** — the chain rule with one extra term $\frac{1}{2}b^2f''$, the price of infinitely wiggly paths.

3 The Poisson process

Crashes do not arrive on a schedule; they arrive *at random times*. The canonical model for random arrivals is the **Poisson process** N_t with intensity λ (events per year):

1. $N_0 = 0$, and N_t counts events up to time t ;
2. independent increments;
3. in a short interval dt : exactly one event with probability λdt , more than one with probability negligible compared to dt .

From these axioms the count over any interval of length t is Poisson distributed:

$$\mathbb{P}(N_t = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad \mathbb{E}[N_t] = \text{Var}(N_t) = \lambda t. \quad (6)$$

Waiting times between events are exponential with mean $1/\lambda$ and memoryless: a market that has not crashed for five years is (in this model) no “safer” than yesterday. Section 5 attaches a random *size* to each arrival to build the jump part of the Merton model.

Part II

Models of Price Motion and Option Pricing

4 Geometric Brownian motion: the Black–Scholes world

4.1 The model

The standard model for an asset price S_t says that *percentage* changes have constant drift and constant noise:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (7)$$

μ is the expected growth rate per year, σ the volatility per year. Percentage (not absolute) changes, because a \$1 move means something different at $S = 10$ than at $S = 1000$.

4.2 Solving it, and the origin of $-\frac{1}{2}\sigma^2$

Apply Itô's lemma (5) to $f(S) = \ln S$ (so $f' = 1/S$, $f'' = -1/S^2$), with $a = \mu S$ and $b = \sigma S$:

$$d \ln S = \left(\frac{\mu S}{S} - \frac{1}{2} \frac{\sigma^2 S^2}{S^2} \right) dt + \frac{\sigma S}{S} dW = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW. \quad (8)$$

Integrating from 0 to t :

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right], \quad \ln \frac{S_{t+\Delta t}}{S_t} \sim \mathcal{N} \left(\left(\mu - \frac{\sigma^2}{2} \right) \Delta t, \sigma^2 \Delta t \right). \quad (9)$$

So under GBM, **daily log-returns are i.i.d. Gaussian**. That is the falsifiable statement the data will judge in Part III. The $-\sigma^2/2$ is not a convention; it is forced by Itô's lemma and reflects the gap between the average of growth rates and the growth rate of the average. The simulator uses (9) directly (the “exact log-Euler” step), so path simulation carries no discretisation error.

4.3 What GBM asserts about the world

No jumps (paths are continuous); constant volatility; Gaussian tails (recall the tail table in §1.2); returns independent day to day. Each assertion can fail in reality; this project focuses on the most dangerous failure for an options desk — the missing jumps — because a delta hedge can, in principle, track a continuous path but can *never* track a gap.

5 The Merton jump-diffusion: a world that can gap

5.1 Construction

Superimpose on GBM a stream of jumps: arrivals from a Poisson process N_t with intensity λ (§3); at each arrival the price is multiplied by e^Y with $Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$ drawn independently. Between jumps the price diffuses as GBM. The SDE is

$$\frac{dS_t}{S_{t-}} = (\mu - \lambda k) dt + \sigma dW_t + (e^Y - 1) dN_t, \quad k = \mathbb{E}[e^Y] - 1 = e^{\mu_J + \sigma_J^2/2} - 1. \quad (10)$$

The value of k uses the lognormal-mean identity $\mathbb{E}[e^Y] = e^{\mu_J + \sigma_J^2/2}$ for Gaussian Y — a computation worth doing once by completing the square in $\int e^y \varphi(y; \mu_J, \sigma_J^2) dy$.

The compensator. Jumps with negative mean would drag the expected growth rate below μ ; the term $-\lambda k dt$ (the *compensator*) subtracts the expected jump contribution so μ remains the total expected growth rate. Without it, comparing “GBM with drift μ ” against “Merton with drift μ ” would compare two different expected returns and the desk experiments would be confounded.

5.2 The distribution of a daily return: a Poisson mixture of Gaussians

Condition on the number of jumps n in the day. Given n , the log-return is Gaussian — diffusion contributes mean $(\mu - \sigma^2/2 - \lambda k)\Delta t$ and variance $\sigma^2\Delta t$; the n i.i.d. jump sizes add mean $n\mu_J$ and variance $n\sigma_J^2$. Weighting by the Poisson probabilities (6):

$$f(x) = \sum_{n=0}^{\infty} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^n}{n!} \varphi\left(x; \left(\mu - \frac{\sigma^2}{2} - \lambda k\right)\Delta t + n\mu_J, \sigma^2\Delta t + n\sigma_J^2\right). \quad (11)$$

A **mixture of Gaussians with different variances is not Gaussian**: the high-variance components dominate far from the centre, producing fat tails, and $\mu_J < 0$ shifts those components left, producing negative skew. This is the entire mechanism by which the extra parameters $(\lambda, \mu_J, \sigma_J)$ buy realistic tails.

6 Options and arbitrage-free pricing

6.1 Contracts and payoffs

A **European call** with strike K and expiry T pays $\max(S_T - K, 0)$; a **put** pays $\max(K - S_T, 0)$. A put is crash insurance: it pays exactly when the market has fallen below K . “Moneyiness” K/S locates the strike relative to spot (an at-the-money (ATM) option has $K \approx S$; an out-of-the-money (OTM) put has $K < S$).

6.2 No-arbitrage and put-call parity

An **arbitrage** is a free lunch — a trade with no initial cost, no risk of loss, and positive probability of gain. Pricing proceeds by ruling these out. The cleanest example: hold (long call, short put, same K, T). At expiry this pays $\max(S_T - K, 0) - \max(K - S_T, 0) = S_T - K$ in every state. A position paying $S_T - K$ for sure must cost its discounted certain value today:

$$C - P = S_0 - Ke^{-rT} \quad (\text{put-call parity}), \quad (12)$$

where r is the risk-free interest rate. Parity is model-free — it holds no matter how S moves — which makes it a perfect *unit test*: our pricer satisfies it to 10^{-10} .

6.3 Delta hedging and the Black–Scholes argument

Why does a *model* give a unique price at all? Black and Scholes’s insight: under GBM the option’s risk can be *eliminated* by trading the underlying. Let $V(S, t)$ be the option value. By Itô’s lemma,

$$dV = \left(V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt + \sigma SV_S dW. \quad (13)$$

Form a portfolio: long the option, short $\Delta = V_S$ shares. Its change is $dV - V_S dS$, and the dW terms *cancel exactly*:

$$dV - V_S dS = \left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt. \quad (14)$$

A riskless portfolio must earn the risk-free rate r (else arbitrage), giving the **Black–Scholes PDE**

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0, \quad (15)$$

whose solution with terminal condition $\max(S - K, 0)$ is the **Black–Scholes formula**

$$C_{BS} = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (16)$$

Φ the standard normal CDF; the put follows from parity (12). Two facts to notice. *First*, μ has vanished: because the hedge removes directional risk, the price cannot depend on expected return — only on volatility. *Second*, the argument needs continuous paths (to rebalance Δ continuously) and a known constant σ ; both fail in a jump world, which is precisely the vulnerability this project quantifies. The hedge ratios used throughout are $\Delta_{\text{call}} = \Phi(d_1)$, $\Delta_{\text{put}} = \Phi(d_1) - 1$, and “vega” $= \partial V / \partial \sigma = S\sqrt{T} \varphi(d_1)$ (needed in §10).

6.4 Risk-neutral pricing in one paragraph

The PDE argument is equivalent to a probabilistic recipe: price as if the world were risk-neutral — replace μ by r in the dynamics, then take the discounted expected payoff, $V = e^{-rT} \mathbb{E}^Q[\text{payoff}]$. The measure Q under which this works is called the *risk-neutral measure*. The practical upshot used later: **option prices reveal the market’s probabilistic model** (under Q) — fitting a model to option prices recovers the dynamics the *market* is pricing, as opposed to fitting returns, which recovers the dynamics the world actually follows (the *physical* measure P).

7 Pricing under Merton: a series of Black–Scholes prices

Under the risk-neutral Merton dynamics, condition on the number of jumps n during the option’s life (probability $e^{-\lambda'T} (\lambda'T)^n / n!$ with $\lambda' = \lambda(1+k)$ — the jump-size tilt $1+k$ arises from the measure change). Given n jumps, the terminal log-price is Gaussian, so each conditional expectation is a Black–Scholes price with adjusted inputs, and the total price is the weighted sum — **Merton’s series**:

$$C_M = \sum_{n=0}^{\infty} \frac{e^{-\lambda'T} (\lambda'T)^n}{n!} C_{BS}(S_0, K, T; r_n, \sigma_n), \quad \sigma_n^2 = \sigma^2 + \frac{n\sigma_J^2}{T}, \quad r_n = r - \lambda k + \frac{n(\mu_J + \sigma_J^2/2)}{T}. \quad (17)$$

Interpretation of the adjustments: n jumps add $n\sigma_J^2$ of log-variance (spread over the life T , hence the $/T$), and shift the risk-neutral growth by their total expected log-size, with the compensator $-\lambda k$ keeping the whole package arbitrage-free. Implementation truncates at 60 terms; since the weights are Poisson, the neglected tail is bounded by $\mathbb{P}(N > 60) < 10^{-40}$ for every parameter set used here. The Merton delta is the same weighted average of Black–Scholes deltas.

Why $C_M > C_{BS}$ at equal diffusive σ . An option payoff is convex. By Jensen’s inequality, spreading the terminal distribution wider (fat tails) raises the expected payoff. The dollar gap $C_M - C_{BS}$, evaluated at calibrated parameters, is the *mispricing per contract* a Black–Scholes desk gives away in a jump world — it will reappear in Part IV as roughly \$10 per contract.

7.1 Implied volatility and the smile

Quoting prices in dollars makes options with different strikes incomparable. The market convention is to quote each option by its **implied volatility**: the unique σ that makes (16) reproduce the price (unique because the BS price is strictly increasing in σ ; we invert numerically with Brent’s root-finder, discarding quotes that violate no-arbitrage bounds). If the world were GBM, every strike would imply the *same* σ : a flat line. Real markets display a **skew/smile**: OTM puts imply much higher volatilities. Under a jump model this is exactly what should happen — crash risk inflates the value of low-strike puts specifically. The smile is thus a diagnostic plot for model error, read directly off market data in §10.3.

Part III

Estimation: Fitting the Models to Real Data

8 Statistical machinery from scratch

8.1 Likelihood and maximum likelihood

Given data $x_1, \dots, x_n$ assumed i.i.d. from a density $f(x; \theta)$ with unknown parameter vector θ , the **likelihood** is the joint density read as a function of θ :

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta), \quad \ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta). \quad (18)$$

The **maximum-likelihood estimator (MLE)** $\hat{\theta}$ maximises ℓ : it is the parameter value under which the observed data were most probable. Under regularity conditions the MLE is consistent (converges to the truth) and asymptotically normal:

$$\hat{\theta} \approx \mathcal{N}(\theta, I(\theta)^{-1}), \quad I(\theta) = -\mathbb{E} \left[\frac{\partial^2 \ell}{\partial \theta \partial \theta^\top} \right] \quad (\text{Fisher information}). \quad (19)$$

In practice I is estimated by the *observed information* — the negative Hessian of ℓ at $\hat{\theta}$, computed numerically — and **standard errors** are the square roots of the diagonal of its inverse. Intuition: the information matrix measures how sharply peaked the likelihood is; a sharp peak means the data pin the parameter down tightly.

8.2 The delta method

If $\hat{\theta}$ has standard error s and we report a transformation $g(\hat{\theta})$, a first-order Taylor expansion gives

$$\text{se}(g(\hat{\theta})) \approx |g'(\hat{\theta})| s. \quad (20)$$

We use this twice: to convert SEs of (daily mean, daily variance) into SEs of annual (μ, σ) , and to convert SEs of log-parameterised Merton parameters (positivity is enforced by optimising $\ln \sigma, \ln \lambda, \ln \sigma_J$) back to the natural scale.

8.3 Hypothesis tests, p-values, and the likelihood-ratio test

A **hypothesis test** confronts a null hypothesis H_0 (e.g. “GBM is the true model”) with data. A *test statistic* is chosen whose distribution under H_0 is known; the **p-value** is the probability, if H_0 were true, of a statistic at least as extreme as observed. Small p-value $\Rightarrow$ the data are implausible under $H_0 \Rightarrow$ reject. Rejecting a true null (false alarm) is a *type I error*, controlled at the significance level (5% throughout); failing to reject a false null is a *type II error*, controlled by test power.

For **nested models** (the smaller obtained from the larger by fixing parameters — here GBM = Merton with $\lambda = 0$), *Wilks’s theorem* says that under H_0

$$LR = 2(\ell_{\text{big}} - \ell_{\text{small}}) \sim \chi_d^2, \quad (21)$$

d = number of restricted parameters (here $d = 3$: λ, μ_J, σ_J). The 95% critical value of χ_3^2 is 7.81.

A boundary caveat, stated honestly. Wilks’s theorem assumes the null lies in the *interior* of the parameter space. Our null $\lambda = 0$ sits on the boundary (λ cannot be negative), and when $\lambda = 0$ the jump-size parameters μ_J, σ_J become meaningless (unidentified). Both issues push the true null distribution of LR below χ_3^2 , so using χ_3^2 is *conservative*: it makes rejecting harder,

not easier. With observed statistics above 1,300 against a critical value of 7.81, no correction could change the verdict — but at small samples this caveat is the difference between careful and careless inference.

8.4 AIC and BIC

Adding parameters always increases the maximised likelihood; information criteria penalise the increase to prevent overfitting:

$$AIC = 2k - 2\ell, \quad BIC = k \ln n - 2\ell, \quad (22)$$

k = number of parameters. Lower is better. BIC's penalty grows with sample size, making it the stricter criterion; a BIC difference above 10 is conventionally called “decisive”. We report both alongside the LR test so the conclusion does not hinge on a single criterion.

8.5 QQ plots

A quantile–quantile plot is the visual counterpart of a goodness-of-fit test. Sort the data $x_{(1)} \leq \dots \leq x_{(n)}$; plot $x_{(i)}$ against the model quantile $F^{-1}((i - \frac{1}{2})/n)$. If the model is right the points lie on the 45° line; systematic S-shaped departures at the ends mean the model's tails are too thin. For Merton, F^{-1} has no closed form; we compute the mixture CDF from (11) on a fine grid and invert by interpolation.

9 Calibration I: the physical measure (what returns actually do)

9.1 GBM: the MLE in closed form

By (9), daily log-returns are i.i.d. $\mathcal{N}(m, v)$ with $m = (\mu - \sigma^2/2)\Delta t$, $v = \sigma^2\Delta t$. Setting $\partial\ell/\partial m = 0$ and $\partial\ell/\partial v = 0$ for the Gaussian log-likelihood yields the sample mean $\hat{m}$ and the (divide-by- n) sample variance $\hat{v}$; mapping back,

$$\hat{\sigma} = \sqrt{\hat{v}/\Delta t}, \quad \hat{\mu} = \frac{\hat{m}}{\Delta t} + \frac{\hat{\sigma}^2}{2}. \quad (23)$$

Exact Gaussian sampling theory gives $\text{Var}(\hat{m}) = v/n$ and $\text{Var}(\hat{v}) = 2v^2/n$; the delta method (20) converts these to SEs for $\hat{\mu}, \hat{\sigma}$.

Observation (drift is unlearnable). With $n \approx 4,500$ days, $\hat{\sigma}$ carries an SE of ± 0.002 — three-decimal precision — while $\hat{\mu}$ carries an SE of $\pm 4.5\%$ *per year*, the same order as the estimate itself. Two decades of daily data cannot tell a 4% drift from a 10% drift. This asymmetry is structural (variance information accumulates much faster than mean information), and it dictates a design choice made later: the desk's P/L is built from *delta-hedged* positions, whose expected value does not depend on μ at all (§11.2).

9.2 Merton: numerical MLE on the mixture density

The log-likelihood is $\sum_i \ln f(x_i)$ with f from (11), truncated at $n \leq 8$ jumps per day — justified because even at the largest fitted λ ($\approx 88/\text{yr}$, $\lambda\Delta t \approx 0.35$), $\mathbb{P}(N > 8) < 10^{-10}$. Three practical complications, each handled explicitly:

1. **Positivity:** optimise $(\mu, \ln \sigma, \ln \lambda, \mu_J, \ln \sigma_J)$ so the search space is unconstrained.

2. **Multi-modality:** diffusion and jumps can partially imitate one another (many small jumps look like diffusion), so the surface has multiple local maxima. We run 8 random restarts of Nelder–Mead (a derivative-free simplex search) and polish the best with BFGS (a quasi-Newton method).
3. **Standard errors:** the observed information — the numerically differentiated Hessian of $-\ell$ at the optimum (central differences) — is inverted, and the delta method maps SEs through the log-links.

Before touching real data the estimator is validated on synthetic data with known parameters (unit test: recovery within sampling error, and $LR > 20$ when jumps are truly present).

9.3 Results and observations

Table 1: Physical-measure MLE on daily log-returns ($\pm$ asymptotic SE). χ^2_3 95% critical value: 7.81.

	SPY (2008–2025)	NIFTY 50 (2007–2026)
n (days)	4,504	4,549
GBM $\hat{\sigma}$	0.2001 ± 0.0021	0.2071 ± 0.0022
Merton $\hat{\sigma}$ (diffusive)	0.0979 ± 0.0039	0.1223 ± 0.0038
Merton $\hat{\lambda}$ (jumps/yr)	87.9 ± 10.6	59.8 ± 8.4
Merton $\hat{\mu}_J$	-0.0026 ± 0.0006	-0.0015 ± 0.0008
Merton $\hat{\sigma}_J$	0.0175 ± 0.0010	0.0207 ± 0.0013
LR vs GBM	1627	1313
ΔBIC (GBM – Merton)	1602	1288

Observation 1 (the verdict). $LR = 1627$ against a critical value of 7.81: the Gaussian model is not marginally worse — it is annihilated. The same on NIFTY 50 ($LR = 1313$): this is a property of equity markets, not of one market. $\Delta BIC > 1,200$ removes any “too many parameters” objection.

Observation 2 (how volatility decomposes). The data reallocate SPY’s 20% Gaussian volatility into a 9.8% smooth diffusion *plus* ≈ 88 jumps per year averaging -0.26% with spread 1.75%. The physical-measure jumps are frequent and small — “jump” here means any discontinuous shock (news, gaps), not only crashes. Rare disasters appear as draws from the σ_J tail: a $3\sigma_J$ jump is a -5.5% day.

Observation 3 (the smoking-gun day). SPY’s worst in-sample day is -11.6% (16 March 2020 — the same date whose option chain is calibrated in §10.3). Under fitted GBM the daily sd is $0.2001/\sqrt{252} = 1.26\%$, making that day a $z \approx -9.2$ event: probability $\sim 10^{-20}$, “once every 3×10^{16} years” by the table in §1.2. The Merton fit assigns it a small but sane probability. Figure 1 shows this in one picture — on a log scale the Gaussian falls away from the empirical tail points by many decades while the mixture tracks them; Figure 2 shows the same as QQ plots.

10 Calibration II: the risk-neutral measure (what options cost)

10.1 Why fit option prices separately

Fitting returns (§9) recovers the world’s actual dynamics (P -measure). Fitting option *prices* recovers the dynamics the market uses to price risk (Q -measure, §6.4) — these can differ, because sellers of crash insurance demand compensation beyond actuarial cost. Both matter to

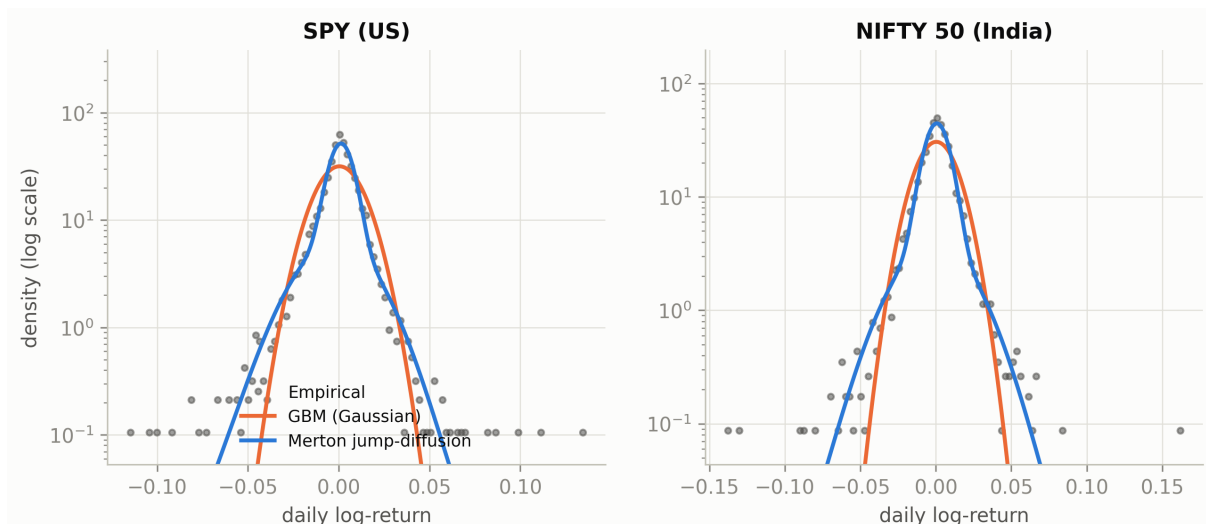


Figure 1: Daily-return densities, log scale: empirical (grey points), fitted GBM (orange), fitted Merton (blue). The log scale is essential: on a linear scale both fits look identical near the centre and the entire story — the tails — is invisible.

a desk: it hedges in the P -world but competes on Q -world prices. We fit Merton to four real SPY chains spanning regimes: 2017-01-03 (calm), 2020-03-16 (COVID crash, VIX 82), 2022-06-13 (bear market), 2025-06-30 (recent).

10.2 The fitting problem, in detail

For each snapshot: take the expiry nearest 90 days to match the desk’s 3-month contract; keep OTM options only (OTM puts below spot, OTM calls above — the liquid side of the smile, and ITM quotes duplicate OTM information via parity); require bid $> \$0.05$ and moneyness in $[0.70, 1.30]$; compute mid-price implied vols (§7.1). Then solve

$$\min_{\sigma, \lambda, \mu, \sigma_I} \sum_i w_i (IV_M(K_i) - IV_{\text{mkt}}(K_i))^2, \quad w_i \propto \exp \left[-\frac{1}{2} \left(\frac{\ln(K_i/S)}{0.15} \right)^2 \right]. \quad (24)$$

Why weight by moneyness? An option’s price sensitivity to volatility is its vega, which peaks at the money; a fixed amount of quote noise (bid–ask bounce) therefore corrupts ATM implied vols least. Gaussian weights centred at ATM encode this. **Why fit IVs rather than prices?** Deep options differ in price by orders of magnitude; a price-space objective would be dominated by the most expensive quotes. IV-space puts all strikes on a common scale. **A speed trick, used honestly:** a first-order Taylor expansion gives $\Delta IV \approx \Delta \text{price} / \text{vega}$, so inside the optimiser we minimise vega-scaled *price* residuals (vega frozen at market IVs — a constant), avoiding a slow numerical inversion at every step; the final reported smile and RMSE are recomputed with exact inversion. Optimisation is multi-start Nelder–Mead with box constraints to keep the simplex out of absurd regions.

10.3 Results and observations

Observation 4 (the smile never believes Black–Scholes). Even in the calmest market at record-low volatility (2017), the fitted smile prices roughly one -10% jump per year. A flat-vol model is wrong on *every* trading day, not just in crises.

Observation 5 (crisis pricing). On 16 March 2020 the market priced $\lambda = 3.68$ crashes/year with mean -31% and spread 19% . These are the parameters used as the “true world” of

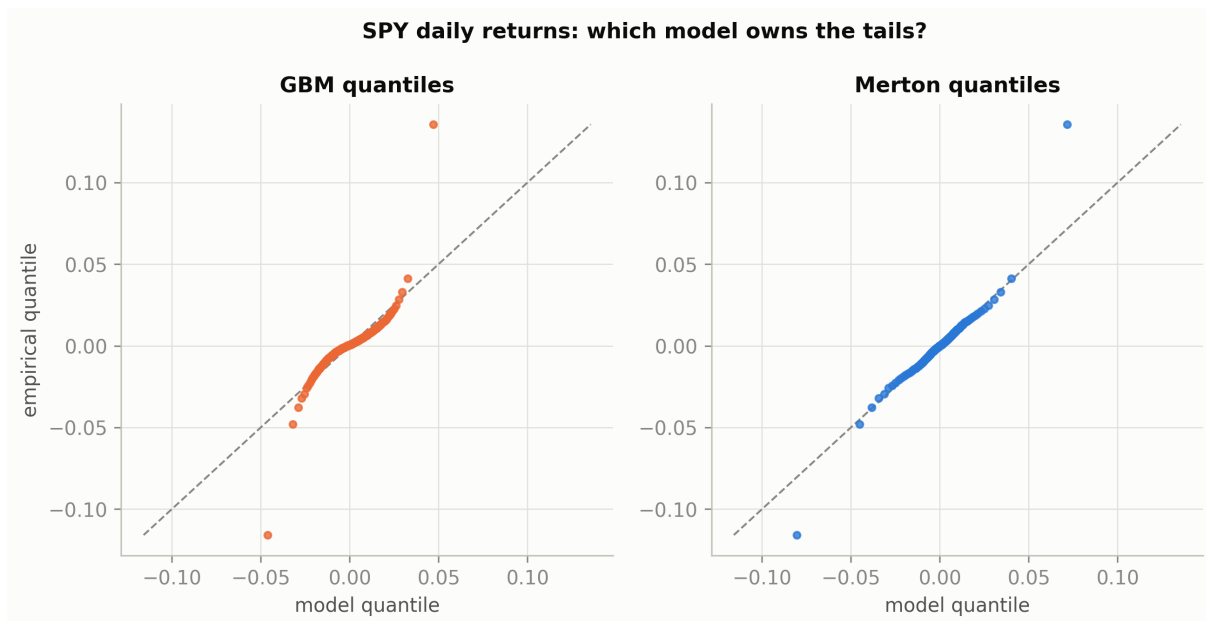


Figure 2: QQ plots of SPY daily returns against each fitted model. Left (GBM): the S-shape means the model's tails are far too thin in both directions. Right (Merton): the points hug the diagonal, including the extreme quantiles.

Table 2: Risk-neutral Merton fitted to four real SPY chains (≈ 90 days to expiry, OTM quotes).

Snapshot	Regime	ATM IV	σ	λ/yr	μ_J	σ_J	RMSE (vol pts)
2017-01-03	calm bull	13.5%	0.068	0.92	-0.106	0.100	1.27
2020-03-16	COVID crash	64.2%	0.116	3.68	-0.308	0.190	0.74
2022-06-13	bear market	30.1%	0.128	2.00	-0.162	0.136	0.67
2025-06-30	recent	14.3%	0.080	0.81	-0.138	0.115	1.05

the steamroller experiment — a jump world market participants demonstrably paid for, not an invention.

Observation 6 (four parameters, forty vol points). The 2020 smile spans ≈ 40 vol points from low-strike puts to high-strike calls; the 4-parameter Merton fit reproduces it to 0.74 vol points RMSE (Figure 3). Parsimonious structure, not curve-fitting: the same four numbers must explain every strike simultaneously.

Observation 7 (P vs Q). Compare Tables 1 and 2: the physical measure shows many small jumps; the risk-neutral measure shows few large ones. The market *overweights* disasters relative to their actuarial frequency — that premium is the compensation demanded for holding crash risk, and it is why the right-model desk in a jump world still earns more than the naive half-spread (Observation 10).

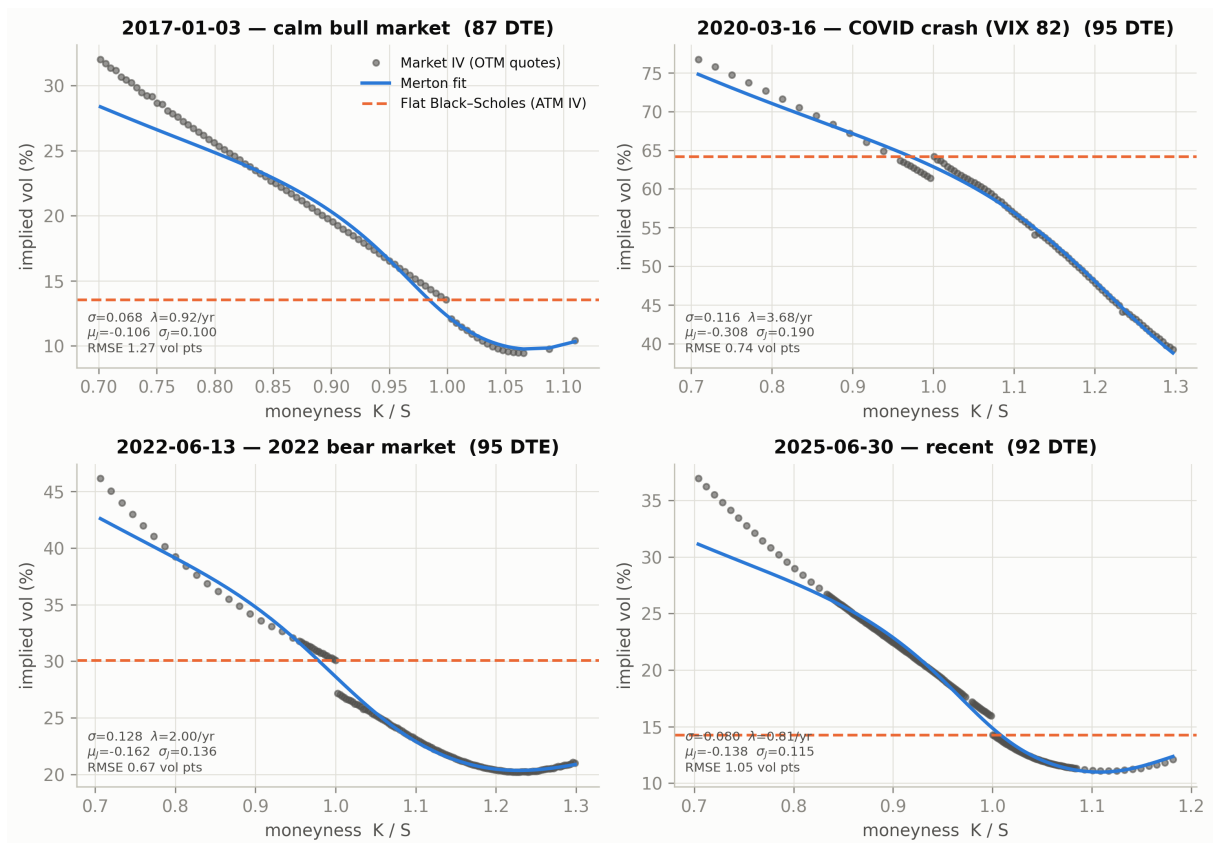


Figure 3: Real SPY implied-vol skews (grey), Merton fits (blue), and the flat smile Black-Scholes implies (orange dashed), across four regimes. The annotation box in each panel reports the fitted parameters and RMSE.

Part IV

The Desk: Market-Making P/L Under Model Risk

11 The simulation, piece by piece

11.1 The quoting cycle

Time advances in ticks of $dt = 1/250$ years ($\approx$ one trading day). At every tick the desk:

1. observes the spot S_t (generated by the *true* model — GBM or Merton — which the desk cannot see);
2. prices a **fresh** option with constant maturity $T = 0.25$ and strike $K_t = S_t$ (rolling at-the-money) using its *believed* model: $V_{\text{mod}} = V(S_t, K_t, T)$;
3. posts bid = $V_{\text{mod}} - s$ and ask = $V_{\text{mod}} + s$ with half-spread $s = \$0.15$;
4. receives client flow (§11.3); each fill is realised immediately (§11.2).

Why a rolling contract at rolling moneyness? A desk is a *business*, not a single bet; quoting the same kind of contract every day keeps its risk profile stationary, so P/L statistics across time are meaningful. This was learned the hard way: an earlier version fixed the strike $K = S_0$ forever, and over multi-year runs the drifting spot pushed the option far out of the money —

shrinking premiums, shrinking risk, and silently breaking the VaR backtest of §15 for the correct model. A validation tool caught a simulation artefact; the fix (roll the strike with spot) is exactly the kind of specification issue model validators exist to find.

11.2 Why a hedged option earns “premium minus fair value”

When a client trades, the contract’s whole life is simulated immediately: the underlying evolves under the *true* model over $[0, T]$ on a 30-step grid, and the desk delta-hedges at each step *using its own model’s delta*, financing at r . The discounted P/L of a short-at-premium- p put position is

$$P/L = p + \sum_{j=0}^{29} e^{-rt_j} h_j (S_{t_{j+1}} - S_{t_j} e^{r\delta t}) - e^{-rT} \max(K - S_T, 0), \quad (25)$$

where h_j is the model-delta hedge held over $[t_j, t_{j+1}]$ (signs flip for bought contracts and for calls). The middle term is the hedge account: you earn the stock move but pay financing on the capital tied up, hence the financing-correct increment $S_{t_{j+1}} - S_{t_j} e^{r\delta t}$.

The key theorem, informally. If the believed model is the true model and rebalancing is continuous, the hedge term exactly cancels the randomness of the payoff (that was the content of the Black–Scholes argument, §6.3), and (25) collapses to the deterministic quantity

$$P/L \rightarrow p - V_{\text{fair}}, \quad (26)$$

independent of the drift μ . Selling at $p = V + s$ therefore banks exactly $+s$; buying at $V - s$ banks $+s$ too. **Spread capture is a theorem — conditional on the model being right.**

Where it breaks, precisely.

- *Discrete hedging* (30 steps, not continuous): the cancellation is approximate; a mean-zero noise per contract survives, shrinking like $1/\sqrt{\text{steps}}$. Visible as the wobble of realised P/L around the expected lines in Figure 4 (left).
- *Wrong fair value*: quotes centre on V_{mod} , so each sale banks $s - (V_{\text{true}} - V_{\text{mod}})$ in true expectation — a systematic, sign-definite leak of about \$10/contract at COVID parameters.
- *Wrong delta + jumps*: a jump moves the price a finite amount in an instant; no rebalancing scheme, however frequent, can follow it. The hedge is simply short the jump. This is why the damage is not just “smaller edge” but fat-tailed losses.
- *Incompleteness*: even the *right-model* desk cannot hedge jumps with the underlying alone; some jump risk is unhedgeable and a small drift exposure survives. Observation 10 quantifies the consequence.

11.3 Client flow: elastic, informed, put-heavy

Each tick, a client may lift the ask (desk sells) with base probability 0.5 and may hit the bid (desk buys) with base probability 0.1 — **put-heavy flow**, matching the real net demand for index crash protection (institutions buy puts to insure portfolios). Crucially, clients are not price-blind. Define the client’s disadvantage in units of half-spread, e.g. for the ask:

$$\text{excess}_{\text{ask}} = \frac{\text{ask} - V_{\text{true}}}{s} - 1 \quad (0 \text{ for a fairly-priced quote}), \quad (27)$$

and scale the fill probability

$$p_{\text{fill}} = \min(0.98, p_0 e^{-\eta \text{excess}}), \quad \eta = 1, \quad (28)$$

with the multiplier capped at $e^2 \approx 7.4\times$ (client arrival is finite even when the desk gives money away). Three regimes fall out naturally: a fairly-priced desk trades at base rates; a *rich* quote

loses business (clients shop elsewhere); a *cheap* quote is hammered by informed flow — **adverse selection**. Without elasticity ($\eta = 0$) the simulator reproduces the textbook toy in which an overpricing desk looks brilliant because clients overpay forever; with it, both directions of mispricing are punished, which drives Observation 11.

11.4 The four scenarios

All share $S_0 = 100$, ATM puts, $T = 0.25$, $r = 2\%$, $\mu = 6\%$, $s = 0.15$, 250 ticks (one year), independent seeds:

Scenario	True world	Desk believes
Matched	GBM, $\sigma = 0.20$ (SPY fit)	Black-Scholes, same σ
Steamroller	Merton, COVID params	Black-Scholes, $\sigma = 0.116$ only
Right model	Merton, COVID params	Merton, true params
Overcautious	GBM, $\sigma = 0.20$	Merton with COVID jumps

(COVID params: $\sigma = 0.116$, $\lambda = 3.68$, $\mu_J = -0.308$, $\sigma_J = 0.19$ from Table 2.) “Steamroller” evokes the trading aphorism about strategies that “pick up pennies in front of a steamroller”: steady small gains punctuated by catastrophic loss. Each scenario runs across hundreds of seeds; a seed fixes the entire random path, so results are exactly reproducible.

12 Results I: what happened, and why

Table 3: Desk experiments across seeds. “Believed edge” is what the desk’s own accounting reports per contract. Realised-edge CI: 95% percentile bootstrap (§16). Negative VaR/ES = the 5% tail is still a profit.

Scenario	Seeds	Believed	Realised edge [95% CI]	Mean P/L	VaR _{5%}	ES _{5%}
Matched	300	+0.15	+0.153 [0.146, 0.159]	+22.8	−9.4	−4.0
Right model	200	+0.15	+0.194 [0.064, 0.324]	+28.3	207.9	289.7
Steamroller	300	+0.15	−10.20 [−10.62, −9.78]	−2497.8	4012.5	4466.9
Overcautious	200	+0.15	−9.08 [−9.23, −8.92]	−1671.1	2044.7	2127.9

Observation 8 (the matched desk delivers its promise, exactly). Realised edge +0.153 with 95% CI [0.146, 0.159] straddling the quoted 0.15: spread capture works, and the residual wobble is the predicted discrete-hedging noise. This row is the *control experiment* — it certifies that the machinery (pricer, hedger, accounting) is sound, so everything that goes wrong elsewhere is attributable to the model mismatch and nothing else.

Observation 9 (belief vs reality, the headline). The steamroller desk books +0.15/contract in its own accounting while realising −10.20. The believed-edge line is not slightly optimistic — it has the wrong *sign*, by a factor of −68. Figure 4 (right) shows the mechanism in time: the teal “belief” line climbs serenely while realised P/L staircases downward through jump losses. Note the mean loss per contract ($\approx \$10.2$) matches the pricing gap $C_M - C_{BS}$ at COVID parameters predicted in §7 — worsened by adverse selection, since informed clients hit the too-cheap ask more often than base rate.

Observation 10 (the right model in a jump world: better, not serene). Realised edge +0.194 > 0.15: the desk collects some jump-risk premium (Observation 7) because clients pay Q -measure prices while jumps realise at P -measure frequency — and, being net short puts, the desk also keeps a small positive drift exposure that continuous-hedging theory would remove but jumps prevent. The price of operating in a jump world is variance: ES_{5%}

of 290 versus the matched desk's -4 — vastly fatter tails *with the correct model*. Jumps are unhedgeable with the underlying alone (an *incomplete market*); knowing the truth shrinks but cannot remove tail risk.

Observation 11 (overpricing is not safety). The overcautious desk — right that the world “might jump”, wrong about this world — loses $-9.08/\text{contract}$. Mechanism: its quotes sit far above true value, so informed clients stop lifting its (rich) ask but hammer its (still-rich) bid, dumping overpriced options on the desk all day. **Model risk is not “being too cheap”;** it is being wrong in either direction when the counterparty knows better. In the price-blind textbook version of this simulator the same desk *prints money*; elasticity (28) is the difference between a toy and an honest model of quoting.

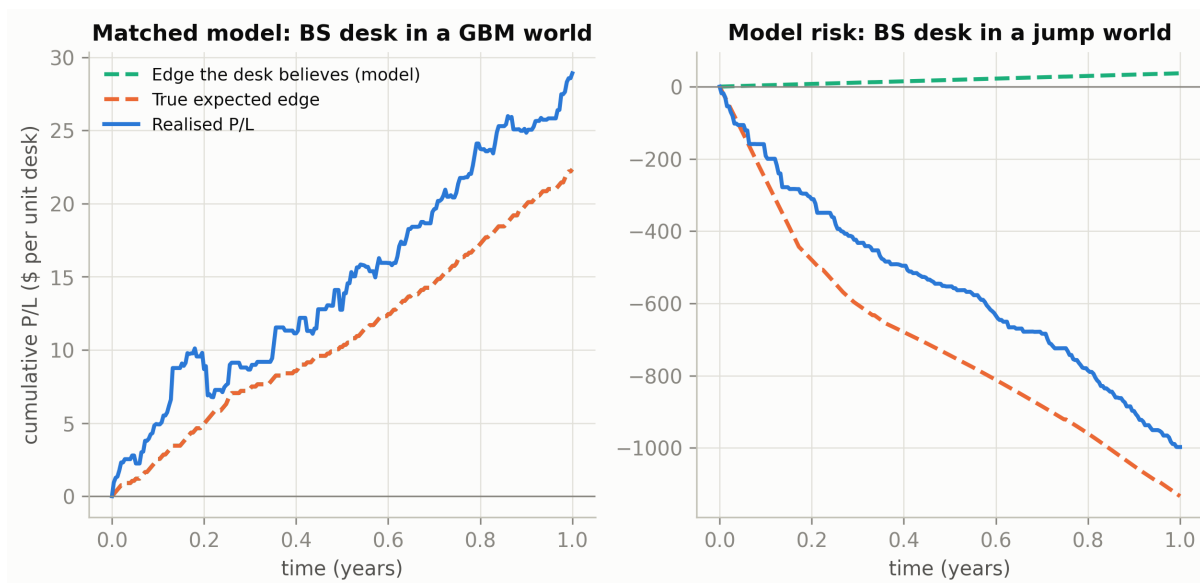


Figure 4: One representative seed. Left, matched: realised P/L (blue) rides the believed (teal dashed) and true (orange dashed) edge lines, which coincide. Right, steamroller: belief and truth diverge from the first fill; realised P/L gaps downward at jumps the hedge cannot follow. The vertical gap between the teal and orange lines *is* model risk, in dollars, accruing per trade.

Part V

Risk Measurement: Pricing the Damage

13 VaR and Expected Shortfall, from scratch

Let L denote the desk's one-year P/L (profit positive) and q_α its α -quantile. Two standard summaries of the left tail:

$$\text{VaR}_\alpha = -q_\alpha(L), \quad \text{ES}_\alpha = -\mathbb{E}[L \mid L \leq q_\alpha(L)]. \quad (29)$$

VaR answers: “what loss is exceeded only with probability α ?” ($\alpha = 5\%$ or 1% conventionally). It is a quantile — intuitive, but it says *nothing about how bad the tail is beyond the threshold*: two desks can share a VaR while one hides a catastrophe behind it. **ES** fixes exactly that by averaging the losses beyond the quantile; it is also mathematically better behaved (*subadditive*: diversifying two books cannot raise their combined ES, which can fail for VaR — the reason banking regulation moved market-risk capital from VaR to ES). Estimation here is empirical:

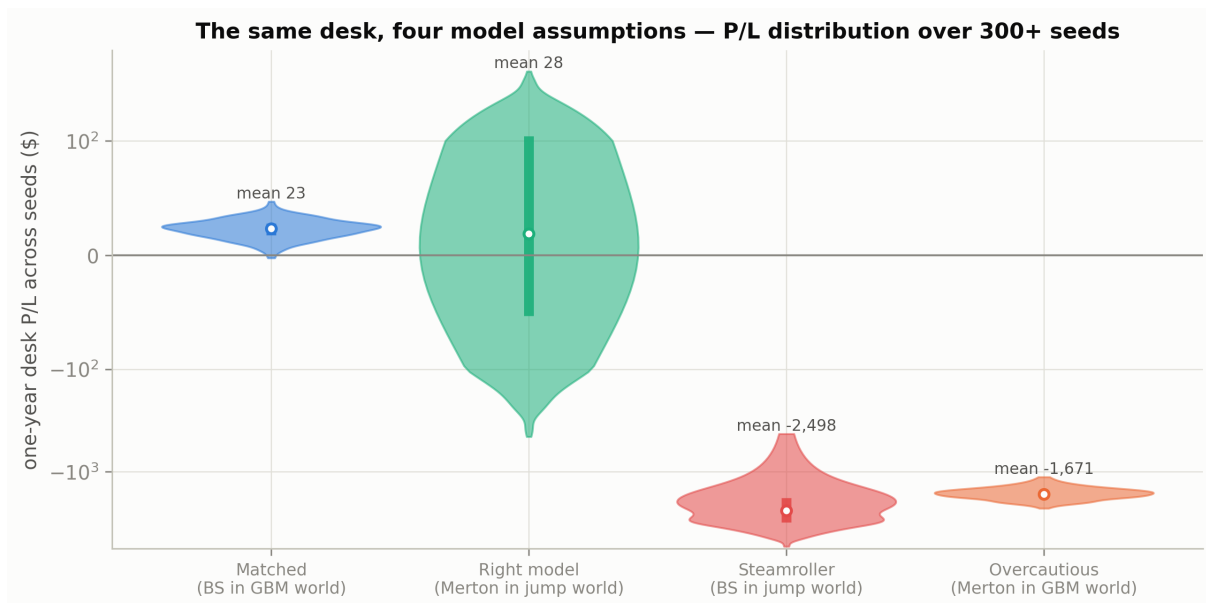


Figure 5: One-year desk P/L across seeds, all four scenarios (symlog scale; violin = full distribution, thick bar = interquartile range, dot = median, annotated mean above). The two mismatched desks are not merely worse on average — their entire distributions sit below zero.

with hundreds of i.i.d. seeds, VaR is the sample quantile and ES the mean of the tail sample. Both appear in Table 3; the steamroller’s $ES_{5\%} = 4,467$ versus the matched desk’s -4.0 is the risk-measurement statement of the whole project.

14 The model-risk premium: a break-even spread search

Spread is the desk’s defence: a wider s banks more per trade. So — can a desk buy safety from a wrong model by quoting wider? Re-run the entire steamroller and matched batches over a grid of half-spreads and estimate the **loss-probability curve** $s \mapsto \mathbb{P}(P/L(s) < 0)$; define the break-even s^* as the smallest s with loss probability $\leq 5\%$ (linear interpolation between grid points). Elastic flow makes this experiment honest: widening quotes also drives clients away, so volume falls as s rises.

Observation 12 (8.8 cents vs \$15+). The matched desk crosses the 5% bar at $s^* = \$0.088$. The steamroller desk’s curve (Figure 6) decays with agonising slowness — at $s = \$15$, more than a hundred times the matched break-even, its loss probability is still $\approx 5.8\%$, so **no half-spread below \$15 clears the bar**. Mechanically: the desk needs s to exceed the $\approx \$10/\text{contract}$ mispricing *plus* compensation for jump variance, but every dollar of extra spread throttles the arrival rate (28), so safety recedes almost as fast as the desk chases it. **The model-risk premium here is not a few cents on the spread; it is the difference between a competitive business and no business.** A desk quoting \$15 half-spreads on a \$10 option has no clients; the only real remedies are a better model or exiting the product.

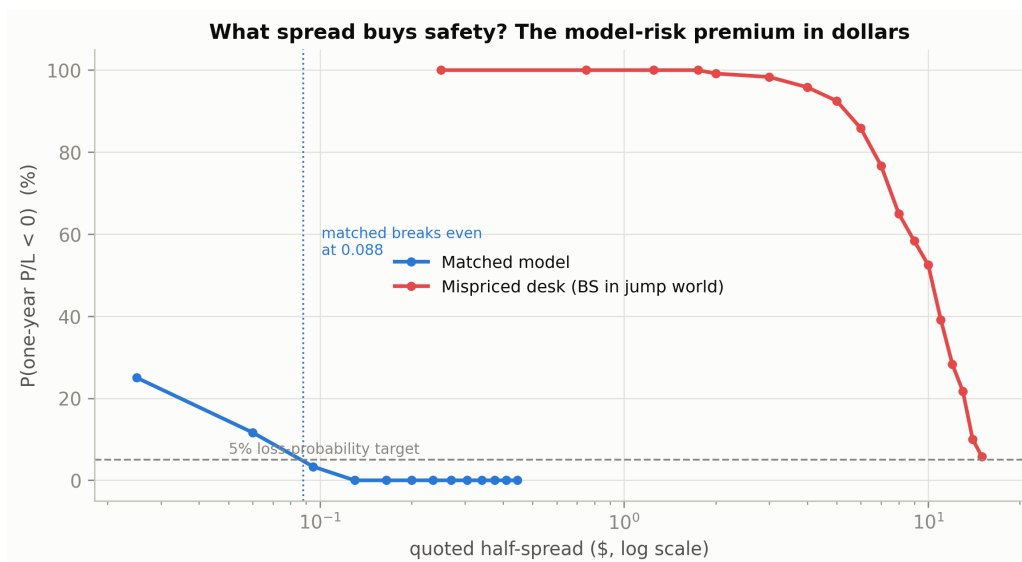


Figure 6: Loss probability vs quoted half-spread (log-scale x). Blue: matched desk, break-even at 8.8 cents. Red: the mispriced desk never reaches the 5% target inside the plotted range.

Part VI

Validation: Catching the Wrong Model from P/L Alone

A model validator does not observe “the true model”; they observe *outcomes*. This part treats the desk’s model as a statistical hypothesis and tests it against realised P/L — the day job of bank model-risk teams (codified in the Federal Reserve’s SR 11-7 guidance). Setup for all tests: a six-year run (1,500 ticks) for statistical power; and every per-tick P/L is divided by S_t/S_0 — **spot-scaling** — because an ATM option’s dollar value is proportional to the spot level, so risk must be measured per unit of notional for the series to be stationary. (This, too, was learned from a failing backtest: unscaled, the matched desk’s P/L variance grows with the drifting spot and a static VaR fails coverage *for the correct model* — a second simulation artefact caught by validation tooling.)

15 VaR backtesting

15.1 The desk forecasts its own risk

Standard practice: the desk computes VaR from *its own* model. Here, the desk simulates its believed world (Black–Scholes/GBM dynamics with its own parameters) over 25 seeds, pools the spot-scaled per-tick P/L increments, and takes the empirical 5% quantile as its one-tick $\text{VaR}_{5\%}$ forecast. Then it lives in the true world, and each tick is scored: an **exception** occurs when the realised loss exceeds the forecast VaR. If the model is right, exceptions should (i) occur on 5% of ticks and (ii) be independent over time. Kupiec tests (i); Christoffersen tests (ii).

15.2 Kupiec’s proportion-of-failures test, derived

Under H_0 the exception indicator is i.i.d. $\text{Bernoulli}(\alpha)$, so the number of exceptions x in n ticks is $\text{Binomial}(n, \alpha)$. The likelihood of the observed sequence under exception probability π is $\pi^x(1 - \pi)^{n-x}$; its MLE is $\hat{\pi} = x/n$. The likelihood-ratio statistic comparing “ $\pi = \alpha$ ” against

“ π free” is

$$LR_{\text{pof}} = -2 \ln \frac{\alpha^x (1 - \alpha)^{n-x}}{\hat{\pi}^x (1 - \hat{\pi})^{n-x}} \stackrel{H_0}{\sim} \chi_1^2 \quad (\text{Wilks, §8.3}). \quad (30)$$

Worked on our data ($n = 1500$, $\alpha = 0.05$, expected 75):

- *Matched desk*: $x = 83$, $\hat{\pi} = 5.53\%$. Plugging in: $LR_{\text{pof}} = 0.87$, $p = 0.351$. The observed excess of 8 exceptions is entirely consistent with chance. **Correct coverage.**
- *Steamroller*: $x = 811$, $\hat{\pi} = 54.1\%$. The model claimed 5%; reality delivered 54% — the desk blows through its own VaR on more than half of all days. LR_{pof} is in the thousands, $p < 10^{-16}$. **Catastrophic rejection.**

15.3 Christoffersen’s independence and conditional-coverage tests

Correct frequency is not sufficient: 75 exceptions all in one month is a broken model with perfect unconditional coverage. Christoffersen models the exception indicator as a two-state Markov chain. Count transitions n_{ij} (from state $i \in \{0, 1\}$ yesterday to j today) and let $\hat{\pi}_0 = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_1 = n_{11}/(n_{10} + n_{11})$ — the exception probability after a calm day vs after an exception day. Independence means $\pi_0 = \pi_1$; the LR test of that restriction is

$$LR_{\text{ind}} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1 - \hat{\pi}_0)^{n_{00}} \hat{\pi}_0^{n_{01}} (1 - \hat{\pi}_1)^{n_{10}} \hat{\pi}_1^{n_{11}}} \stackrel{H_0}{\sim} \chi_1^2, \quad (31)$$

and the **conditional coverage** statistic $LR_{\text{cc}} = LR_{\text{pof}} + LR_{\text{ind}} \sim \chi_2^2$ tests both properties at once.

Observation 13 (coverage). Matched: Kupiec $p = 0.351$, CC $p = 0.619$ — the correct model passes both, which certifies the tests’ false-alarm behaviour on healthy data. Steamroller: both $p < 10^{-16}$.

Observation 14 (clustering is the fingerprint of jumps). The steamroller’s exceptions do not arrive uniformly: Figure 7 shows them bunched into episodes — a jump craters the P/L and the elevated loss regime persists while the desk keeps quoting through it, so $\hat{\pi}_1 > \hat{\pi}_0$. Clustered exceptions tell the validator not only *that* the model is wrong but *what kind* of dynamics it is missing: state-dependent risk (jumps, volatility regimes) rather than a mere level miscalibration of VaR.

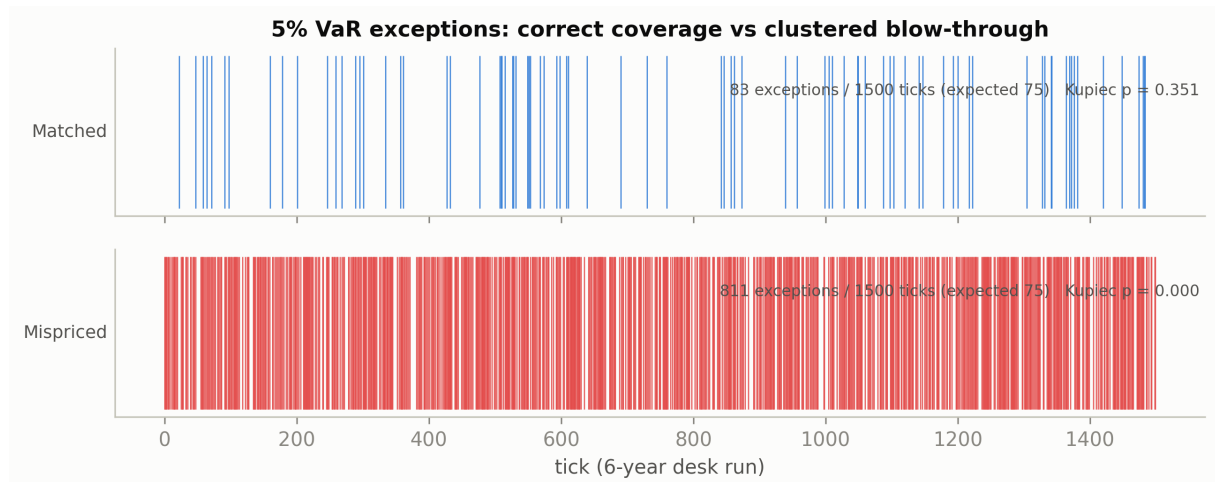


Figure 7: 5% VaR exception timelines over six years. Top (matched): 83 exceptions vs 75 expected, scattered. Bottom (steamroller): 811, clustered into jump episodes.

16 The bootstrap: honest error bars on realised edge

The problem. “The desk averaged +0.153 per contract” — is that meaningfully different from the promised 0.15? From zero? Any answer needs the sampling distribution of the mean, and the P/L distribution is skewed and fat-tailed, so Gaussian formulas are suspect.

The plug-in principle. The bootstrap replaces the unknown population by the sample itself: draw n values *with replacement* from the observed per-seed edges, compute the mean, repeat $B = 10,000$ times; the spread of those B means estimates the sampling variability of the original mean. The **percentile interval** takes the 2.5th and 97.5th percentiles of the bootstrap means as a 95% CI. Validity here is clean: seeds are i.i.d. by construction, which is exactly the bootstrap’s core assumption. A one-sample t -test of H_0 : mean = 0 is reported alongside as a cross-check.

Observation 15. Matched: +0.153, CI [0.146, 0.159] — excludes zero decisively, contains the promised 0.15: the desk delivers its quoted edge and the theory of §11.2 is confirmed *with error bars*, not by eyeball. Steamroller: −10.20, CI [−10.62, −9.78] — significantly, catastrophically negative. Right-model: CI [0.064, 0.324] — positive but wide, quantifying how much harder it is to measure edge in a fat-tailed world (similar seed count, several times the uncertainty).

17 CUSUM: the early-warning monitor

17.1 The problem: detect a change, fast, without crying wolf

The backtests above are retrospective — they need a long sample. A desk bleeding edge should be caught in *weeks*. This is sequential change detection: monitor a stream $X_1, X_2, \dots$ (here: spot-scaled per-trade edge residuals, $X_t = \text{realised edge} - \text{promised edge}$) and raise an alarm as soon as its mean has shifted down, while rarely alarming if nothing changed. The two error rates trade off: a hair-trigger monitor detects fast but false-alarms constantly; a sluggish one never cries wolf and never warns.

17.2 The CUSUM statistic

Page’s one-sided CUSUM accumulates evidence of under-delivery but never lets credit build up (the $\max(0, \cdot)$ clamp — past overperformance must not mask a current bleed):

$$C_0 = 0, \quad C_t = \max(0, C_{t-1} + (m - X_t) - k), \quad \text{alarm when } C_t > h. \quad (32)$$

Here m is the promised edge, k an *allowance* (slack absorbing normal noise; we use $k = 0.5\hat{\sigma}$, the classic choice — tuned to half the shift one wants to detect, a rule inherited from the statistic’s sequential-likelihood-ratio derivation), and h an *alarm threshold* ($8\hat{\sigma}$) controlling the false-alarm rate: larger h means rarer false alarms but slower detection (formalised by *average run lengths*: the mean time to a false alarm on a healthy stream, ARL_0 , versus the mean detection delay after a real shift, ARL_1). The noise scale $\hat{\sigma}$ is estimated on a **burn-in window** which is then *excluded from monitoring* — you do not raise alarms on the data used to set the control limits, a rule inherited from industrial quality control.

Observation 16 (detection in three weeks, zero false alarms). On the steamroller desk the CUSUM crosses h at **trade 83 of 1,467** — roughly three weeks into a six-year run, long before the equity curve is visually diagnostic and *years* before a Kupiec test would have a meaningful sample. On the matched desk, the statistic breathes below threshold for all 903 trades: no false alarm (Figure 8). The pairing matters: a monitor is only as good as

its false-alarm behaviour on healthy systems, which is why both curves are shown on the same limits.

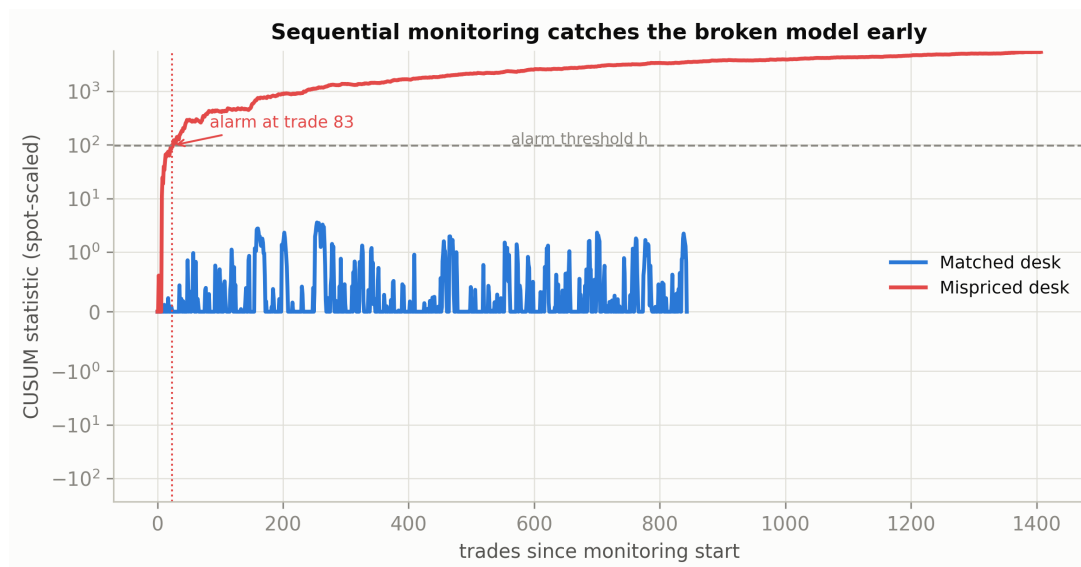


Figure 8: One-sided CUSUM on spot-scaled edge residuals (symlog y). Blue: matched desk, never alarms. Red: mispriced desk, crosses the threshold at trade 83.

The validator’s playbook, assembled. Coverage tests (§15) ask “is the model’s risk forecast honest?”; clustering tests ask “is the risk *dynamics* right?”; bootstrap inference (§16) asks “is the claimed edge real?”; CUSUM (§17) asks all of it *continuously, in near-real-time*. Together they detect the wrong model from outcomes alone — no access to “truth” required — with verified false-alarm control. That is the entire logic of model risk management, demonstrated end to end on one desk.

Part VII

Closing Material

18 Limitations, stated plainly

- **Stylised microstructure.** One contract quoted at a time, Bernoulli arrivals, reduced-form elasticity (28) — no order book, no strategic clients, no inventory limits.
- **Fixed truth within runs.** Each experiment’s true world is a fixed-parameter Merton or GBM; real model risk also includes forms neither model spans (stochastic volatility, regime switching) and parameters that drift.
- **Boundary null** in the LR test (§8.3) — conservative here, and decisively overwhelmed at these statistic sizes.
- **Incomplete-market economics compressed.** Jump-risk premia (P – Q gaps) are real and only partially disentangled from hedging residuals in Observation 10; a full treatment would separate them.
- **Data vintage.** Option chains end December 2025; NIFTY options are not in the free mirror (the repo ships a fetch script to extend both).

None of these threaten the qualitative conclusions; each is a concrete extension path.

19 Reproducibility and repository map

Every number and figure in this document regenerates from two commands:

```
python scripts/run_experiments.py → python scripts/make_figures.py
```

Seeds are fixed throughout; 15 unit tests (`python -m pytest tests/`) pin put–call parity (10^{-10}), deltas against finite differences, Merton $\rightarrow$ BS convergence as $\lambda \rightarrow 0$, MLE parameter recovery on synthetic data, the matched desk’s realised edge, and the acceptance/rejection/alarm behaviour of every validation test on synthetic series with known properties. The public web-site (`docs/`) contains an interactive JavaScript port of the full engine — seeded RNG, Merton series pricer, hedged-fill realisation, elastic flow — so every experiment in this document can be re-run in a browser. Data provenance is documented file-by-file in `data/README.md`.

Glossary

Adverse selection Losing systematically to better-informed counterparties; here, mispriced quotes getting picked off.

ATM/OTM/ITM At/out-of/in-the-money: strike at, unfavourable to, or favourable to current spot.

Delta Sensitivity of option value to spot, $\partial V / \partial S$; the hedge ratio.

Drift (μ) Expected growth rate of the underlying per year.

Edge Expected profit per trade; for a market maker, the half-spread.

ES (Expected Shortfall) Average loss beyond the VaR quantile.

Half-spread (s) Half the gap between ask and bid.

Implied volatility The σ that makes Black–Scholes match a price.

MLE Maximum-likelihood estimator.

Moneyness K/S ; locates strike relative to spot.

P vs Q measure Real-world vs risk-neutral (pricing) probabilities.

Risk-free rate (r) Return on riskless lending; the discounting rate.

VaR Loss threshold exceeded with probability α .

Vega Sensitivity of option value to volatility.

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